



Small Language Models (SLMs)

Small, yet incredibly capable

In 2024, Small Language Models (SLMs) have become integral to mobile and IoT ecosystems. Their lightweight architecture and localised capabilities cater to applications that demand low latency and minimal energy consumption, making them ideal for offline and on-the-go use cases. Here are some of the standout applications:

Microsoft Translator Lite: Real-time multilingual translations offline

Microsoft Translator Lite leverages compact SLMs to provide real-time translations without requiring an internet connection. In 2024, the application reached over 70 million monthly active

users globally, driven by its inclusion of support for emerging languages like Swahili and Tamil. Microsoft reported that Translator Lite has improved global productivity by reducing communication barriers for over 40% of small-to-medium enterprises using the tool in regions with limited internet access. With support for over 50 languages, including emerging regional dialects, the tool caters to frequent travellers, remote workers, and multilingual professionals. According to Microsoft's internal research, the app has reduced data usage by 80%, which is crucial in regions where mobile data is expensive or unreliable. This innovation addresses the persistent challenge of ensuring accessibility across language barriers while minimising data